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Moveable chaff pan for agricultural vehicle

Abstract

A chaff pan assembly for an agricultural vehicle includes a chaff pan that is moveable between raised and lowered positions. A stationary bracket is mounted to a fixed point on the agricultural vehicle. A linkage interconnects the chaff pan to the stationary bracket and guides movement of the chaff pan between the raised and lowered positions. A locking member is moveably connected to the stationary bracket and is moveable between a locked position, in which the locking member is engaged with one arm of the linkage to prevent movement of the one arm of the linkage, and an unlocked position, in which the locking member is disengaged from the one arm of the linkage to permit movement of the one arm of the linkage.

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Background/Summary

FIELD OF THE INVENTION

(1) This invention relates generally to agricultural combines, and more particularly, to a foldable chaff pan that is positioned between a cleaning system and a chopper and/or spreader of an agricultural combine.

BACKGROUND OF THE INVENTION

(2) As is described in U.S. Patent App. Pub. No. 20030100353 and 20110053667, which are each incorporated by reference in its entirety and for all purposes, it is known to mount a pivotable chaff pan to an oscillating chaffer rail, frame or other enclosure of a combine harvester.

(3) Described herein is a cost-effective and convenient arrangement for locking a chaff pan, which is connected to an oscillating chaffer rail, in both a raised position and a lowered position of the chaff pan. In the raised position, the chaff pan is maintained in a position to pass chaff to a

spreader/chopper. In the lowered position, the chaff pan is positioned to windrow the chaff, and thereby bypass the spreader/chopper.

(4) This description of the background is provided to assist with an understanding of the following explanations of exemplary embodiments, and is not an admission that any or all of this background information is necessarily prior art.

SUMMARY

(5) In a first exemplary aspect, a chaff pan assembly (27) for an agricultural vehicle (12) is provided. The chaff pan assembly includes a chaff pan (28) that is moveable between a raised position and a lowered position; a stationary bracket (65) that is configured to be mounted to a fixed point (22) on the agricultural vehicle (12); and a linkage (52, 58, 60) that interconnects the chaff pan (28) to the stationary bracket (65) and guides movement of the chaff pan (28) between the raised and lowered positions. The chaff pan (28) assembly further comprises a locking member (67) that is moveably connected to the stationary bracket (65) and is moveable between a locked position, in which the locking member (67) is engaged with one arm (60) of the linkage to prevent movement of the one arm (60) of the linkage, and an unlocked position, in which the locking member (67) is disengaged from the one arm (60) of the linkage to permit movement of the one arm (60) of the linkage. The moveable locking member (67) is moveable to the locking position in both the raised and the lowered positions of the chaff pan (28) to prevent movement of the one arm (60) of the linkage in both the raised and lowered positions of the chaff pan (28).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of inventions will now be described, strictly by way of example, with reference to the accompanying drawings, in which:

(2) FIG. 1 is a simplified, side elevational representation of a rear end of an agricultural combine including a foldable oscillating chaff pan according to the present invention, wherein the chaff pan is shown in a raised position.

(3) FIG. 2 is another view of the rear end of the agricultural combine, wherein the chaff pan is shown in a lowered position.

(4) FIG. 3 depicts an isometric view of a lock assembly for connecting the chaff pan to the frame of the combine.

(5) FIGS. 4A and 4B depict side elevation and cross-sectional views, respectively, of a portion of the lever arm and lock assembly, wherein the lever arm is shown in the raised position (corresponding to the raised position of the chaff pan shown in FIG. 1) and the lock assembly is shown in the locked position. FIG. 4B is taken along the lines 4B-4B in FIG. 3.

(6) FIGS. 5A and 5B depict a side elevation and cross-sectional views, respectively, of the lever arm (end portion shown) and lock assembly, wherein the lever arm is shown in the lowered position (corresponding to the lowered position of the chaff pan shown in FIG. 2) and the lock assembly is shown in the locked position.

(7) FIGS. 6A-6F depict a movement sequence of the lever arm (end portion shown) and the lock assembly from the raised and locked position (FIG. 6A) to the lowered and locked position (FIG. 6D) and back toward the raised position. The lever and the lock assembly are shown in cross-section in FIGS. 6A-6F, like FIG. 4B.

DETAILED DESCRIPTION OF THE DRAWINGS

(8) In the figures, like reference numerals refer to the same or similar elements.

(9) The terms “forward,” “rearward,” “left,” and “right”, when used in connection with the agricultural harvester (e.g. combine) and/or components thereof are usually determined with reference to the direction of forward operative travel of the combine, but they should not be

construed as limiting. The terms “longitudinal” and “transverse” are determined with reference to the fore-and-aft direction of the agricultural combine and are also not to be construed as limiting. (10) The drawings depict one or more exemplary and non-limiting implementations in accordance with the present concepts. The examples are shown in conjunction with an agricultural combine harvester, but have applicability in any similar agricultural vehicle, such as a windrower or other equipment.

(11) FIG. 1 shows a rear end **10** of a self-propelled agricultural combine **12**. Combine **12** includes a threshing section (not shown) for separating grain and chaff from straw, stalks, and other residue. After separation from the straw, stalks, and other residue by the threshing system, the grain is separated from chaff by cleaning apparatus **16**. Cleaning apparatus **16** includes at least one generally horizontal chaffer or sieve (not shown) supported on front-to-rear extending chaffer rails **18** (one shown) located on opposite sides of the chaffer or sieve, the chaffer or sieve including apertures therethrough adjustable in size for allowing passage of grains of a selected maximum size therethrough, while preventing the larger chaff from passing therethrough, for separating the grain from the chaff in the well-known conventional manner. Cleaning apparatus **16** is oscillated in one or more generally horizontal directions, as denoted by arrows B, by oscillating apparatus (not shown) to facilitate the separation of the grain from the chaff, also in the well-known conventional manner.

(12) A chaff pan assembly **27** generally includes a chaff pan **28** that is movably connected to a lock assembly **66** by a linkage. Lock assembly **66** is mounted to the frame **22** of combine and includes a locking member that enables locking/unlocking of the chaff pan **28** in either a raised or a lowered position, and the linkage guides movement of the chaff pan **28** between the raised and lowered positions.

(13) At the outset it is noted that although the chaff pan **28** may be described herein as locked in either a raised or lowered position, it should be understood that the chaff pan is not prevented from oscillating movement in those locked positions.

(14) Referring now to the individual components of chaff pan assembly **27**, chaff pan **28** that is attached or mounted to chaff rails **18** of cleaning apparatus **16** so as to oscillate therewith and importantly so as to be pivotable relative thereto between a raised position (FIG. 1) and a lowered position (FIG. 2). In the raised position, chaff pan **28** directs the flow of chaff as represented by arrows D toward rotary device **14** (i.e., a chopper or spreader) for further processing. In the lowered position, chaff pan **28** directs the flow of chaff downward onto the ground in a windrow.

(15) The linkage includes an arm **52** is fixedly connected to the side of pan **28**. A pivot joint **54** pivotally connects arm **52** to a bracket **50** that is fixedly connected to the chaffer rail **18**. Arm **52** has a first elongated portion that is connected to pan **28**, and a second shorter portion that is connected to bracket **50** by pivot joint **54**. One end of a shaker arm **58** is pivotally connected to arm **52** by pivot joint **47**. An opposite end of shaker arm **58** is pivotably connected to a free end of a lever arm **60** by pivot joint **48**. Lever arm **60** has an opposite end that is pivotally connected to frame **22** by a lever arm pivot **62** located on a lock assembly **66** fixedly mounted to frame **22**. Further details of arm **60** will be described with reference to the lock assembly **66**.

(16) Arms **52**, **58** and **60** along with pivot joints **47**, **48** and **54** may be considered as forming a pivotable three-bar linkage. The pivot joints at **47**, **48** and **54** also enable oscillating movement of pan **28** when pan **28** is maintained in either a raised or lowered position. It is noted that mounting the linkage to the frame **22** of the combine distributes the weight of the chaff pan **28** onto the frame **22** of the combine in a desirable way as opposed to the cleaning section of the combine carrying the weight of the chaff pan **28**.

(17) Although one side of chaff pan assembly **27** is shown in FIGS. 1 and 2, it should be understood that the same components are positioned on the other side of chaff pan assembly.

(18) Lock assembly **66** and the end of lever arm **60** that is connected to lock assembly **66** are best

shown in FIGS. 3-4B. Lock assembly **66** generally comprises a stationary bracket **65** having a mounting surface **55** that is fixedly coupled to frame **22** by fasteners, and a locking member **67** that is pivotably (or otherwise moveably) coupled to bracket **65** by a rod **68** that passes through registered openings in both locking member **67** and bracket **65**.

(19) The end of lever arm **60**, which is connected to lock assembly **66**, generally includes (i) two elongated plates **61**, (ii) a winged mounting portion **57** extending from the plates **61** that is positioned between opposing sides of bracket **65** and pivotably mounted to those sides by the pivot joints **62** (one shown), and (iii) a transverse rod **64** passing through the plates **61**. Rod **64** forms a bearing surface, as will be described later. A tab **59** extends from the winged mounting portion **57** to an elevation beneath the plates **61**. At least a portion of tab **59** is positioned between an axis passing through rod **64** and an axis passing through pivot joint **62**. Tab **59** also forms a bearing surface, as will be described later. Also, yet another bearing surface **56** projects from the free end of lever arm **60** for interacting with locking member **67**, as will be described hereinafter. Thus, arm **60** has three different bearing surfaces **64**, **59** and **56** for preventing rotating of the arm **60** in at least one rotational direction, the purpose of each of which will be described later.

(20) Stationary bracket **65** is a one-piece unitary sheet metal or cast component having two planar mounting surfaces/portions **55** (for mounting to frame **22**, as described above), legs **51** longitudinally depending from each mounting surface **55**, and a shoulder **53** extending transversely between and interconnecting the legs **51**. A transversely extending rod **68** is mounted to openings disposed in opposing legs **51** of bracket **65** by bearings **69**. On each leg **51**, a cutout, recess or surface **49** is formed having two end-stop surfaces **49a** and **49b**, the purpose of which will be described later.

(21) Locking member **67** includes an opening, recess or channel in which rod **68** is disposed. Locking member **67** is configured to pivot about rod **68**. Locking member **67** has an open-ended cutout **71** formed in a central region thereof having two opposing bearing surfaces **71a** and **71b**. The lower end of locking member **67** forms a user-accessible handle for moving locking member **67** between its locked and unlocked positions.

(22) A spring-loaded and resilient element **72** has one end that is fixed to locking member **67** and an opposite end that is mounted (or fixed) to bracket **65** by one or more fasteners. Resilient element **72** biases locking member **67** in the clockwise direction (as viewed in FIG. 4B) and toward the locking position shown in FIG. 4B (as well as FIG. 5B). Resilient element **72** may be in the form of a unitary sheet spring steel, for example. Resilient element **72** could also be a tension or compression spring, if so desired.

(23) Turning now to FIGS. 4A and 4B, in the raised position of chaff pan **28**, lever arm **60** is maintained in the raised and locked position (as shown in those figures). More particularly, in the raised and locked position of arm **60**, rod **64** is positioned against end stop surface **49a** to prevent rotation of arm **60** about pivot joint **62** in the clockwise direction (as viewed in FIG. 4A). And, bearing surface **56** of lever arm **60** bears against bearing surface **71a** of locking member **67** to prevent rotation of arm **60** in the counter clockwise direction (as viewed in FIG. 4B) about pivot joint **62**. Thus, lever arm **60** is prevented from pivoting about pivot joint **62**.

(24) As a result of pivot joints **47**, **48** and **54**, chaff pan **28** is capable of oscillating along with chaff rail **18** even when arm **60** is maintained in either a raised or a lowered position and regardless of whether arm **60** is maintained in the locked position by locking member **67**.

(25) Turning now to FIGS. 5A and 5B, in the lowered position of chaff pan **28**, lever arm **60** is maintained in the lowered and locked position (as shown in those figures). More particularly, in the lowered and locked position of arm **60**, rod **64** is positioned against end stop surface **49b** to prevent rotation of arm **60** about pivot joint **62** in the counterclockwise direction (as viewed in FIG. 5A). And, tab **59** of lever arm **60** bears against bearing surface **71b** of locking member **67** to prevent rotation of arm **60** about pivot joint **62** in the clockwise direction (as viewed in FIG. 5B). Thus, lever arm **60** is prevented from pivoting about pivot joint **62** in the lowered and locked position. An

exemplary sequence of movements between the raised/locked and the lowered/locked positions will be described hereinafter.

(26) Turning now to FIGS. 6A-6F, starting from the raised and locked position of arm 60 of FIG. 6A, which corresponds to the raised position of chaff pan 28 shown in FIG. 1, as noted above, arm 60 is locked in the raised position as was described above with reference to FIGS. 4A and 4B. Turning now to FIG. 6B, to release arm 60 and chaff pan 28 from the raised and locked position, locking member 67 is moved in the direction of the arrow in FIG. 6B against the bias of resilient element 72, thereby releasing (i.e., disengaging) bearing surface 56 from locking member 67 as shown. Movement of locking member 67 is preferably performed manually, but could be performed in an automated fashion by solenoids controlled in a cab of the combine.

(27) Turning now to FIG. 6C, to move arm 60 (as well as chaff pan 28) to the lowered position, arm 60 is rotated downwards and in the counterclockwise direction about pivot joint 62, as depicted by the arrow in that figure. Alternatively, chaff pan 28 may be rotated downwardly, which would cause arm 60 to rotate downwardly. The spring-loaded locking member 67 may be released as soon as bearing surface 56 clears the cutout 71 of locking member 67. Upon releasing locking member 67, bearing surface 56 can ride along the flat outer surface of locking member 67, and tab 59 can also ride along the outer surface of locking member 67 as the arm 60 continues to be rotated in the counter clockwise direction. Turning now to FIG. 6D, arm 60 is rotated further in the counterclockwise direction until the cutout 71 springs over tab 59 and the tab 71 becomes positioned within cutout 71. The resilient element 72 rotates locking member 67 so that bearing surface 71b engages against the bottom side of tab 59. At this time, arm 60 is maintained in the lowered and locked position, corresponding to the lowered position of chaff pan 28. The lowered and locked position of arm 60 was described previously in relation to FIGS. 5A and 5B. It should be understood that chaff pan 28 pivots to its lowered position under its own weight as arm 60 is moved to its lowered position, as guided by the above-described linkage.

(28) Turning now to FIG. 6E, to release arm 60 and chaff pan 28 from the lowered and locked position, locking member 67 is moved in the direction of the arrow in FIG. 6E against the bias of resilient element 72, thereby releasing tab 59 from bearing surface 71b. Turning now to FIG. 6F, to move arm 60 (as well as chaff pan 28) to the raised position, arm 60 is rotated upwards and in the clockwise direction about pivot joint 62, as depicted by the arrow of that figure. Spring-loaded locking member 67 may be released as soon as tab 59 clears the cutout 71 of locking member 67. Upon releasing locking member 67, bearing surface 56 can ride along the flat outer surface of locking member 67. Arm 60 is rotated further in the clockwise direction until the cutout 71 springs over the bearing surface 56 of arm 60 due to the spring force of resilient element 72. At that time, arm 60 is maintained in the raised and locked position shown in FIG. 6A. It should be understood that chaff pan 28 pivots to its raised position as arm 60 is moved to its raised position, as guided by the above-described linkage.

(29) It should be understood that the movement sequence described above is not limited to any particular step of sequence of steps.

(30) The present disclosure describes a number of inventive features and/or combinations of features that may be used alone or in combination with each other or in combination with other technologies. The embodiments described herein are all exemplary, and are not intended to limit the scope of the claims. It will also be appreciated that the inventions described herein can be modified and adapted in various ways, and all such modifications and adaptations are intended to be included in the scope of this disclosure and the appended claims.

Claims

1. A chaff pan assembly for an agricultural vehicle, said chaff pan assembly comprising: a chaff pan that is moveable between a raised position and a lowered position; a stationary bracket that is

configured to be mounted to a fixed point on the agricultural vehicle; a linkage that interconnects the chaff pan to the stationary bracket and guides movement of the chaff pan between the raised and lowered positions; and a locking member that is moveably connected to the stationary bracket and is moveable between a locked position, in which the locking member is engaged with one arm of the linkage to prevent movement of the one arm of the linkage, and an unlocked position, in which the locking member is disengaged from the one arm of the linkage to permit movement of the one arm of the linkage, wherein the moveable locking member is moveable to the locking position in both the raised and the lowered positions of the chaff pan to prevent movement of the one arm of the linkage in both the raised and lowered positions of the chaff pan.

2. The chaff pan assembly of claim 1, wherein the chaff pan is configured to be connected to an oscillating chaff rail, and the linkage includes pivot joints that enable oscillating motion of the chaff pan in both the raised and lowered positions of the chaff pan.

3. The chaff pan assembly of claim 1, wherein, in the raised position of the chaff pan, the one arm of the linkage has a first rotational position, and in the lowered position of the chaff pan, the one arm of the linkage has a second rotational position that differs from the first rotational position.

4. The chaff pan assembly of claim 1, wherein the linkage is a three-bar linkage and the one arm of the linkage forms one of the bars of the three-bar linkage.

5. The chaff pan assembly of claim 1, further comprising a resilient member connected to the locking member for urging the locking member to the locked position.

6. The chaff pan assembly of claim 1, wherein the locking member is moveable to the unlocking position in both the raised and the lowered positions of the chaff pan to permit movement of the one arm of the linkage in both the raised and lowered positions of the chaff pan.

7. The chaff pan assembly of claim 1, wherein the one arm comprises three different bearing surfaces, wherein in a raised and locked position of the chaff pan, a first bearing surface of the three different bearing surfaces is engaged with a first mating bearing surface of the stationary bracket to prevent movement of the one arm in a first direction, and a second bearing surface of the three different bearing surfaces is engaged with a first mating bearing surface of the locking member to prevent movement of the one arm in a second direction that is opposite the first direction.

8. The chaff pan assembly of claim 7, wherein in a lowered and locked position of the chaff pan, the first bearing surface of the three different bearing surfaces is engaged with a second mating bearing surface of the stationary bracket to prevent movement of the one arm in the second direction, and a third bearing surface of the three different bearing surfaces is engaged with a second mating bearing surface of the locking member to prevent movement of the one arm in the first direction.

9. The chaff pan assembly of claim 8, wherein the first and second mating bearing surfaces of the stationary bracket are defined by a channel formed in the stationary bracket.

10. The chaff pan assembly of claim 8, wherein the first and second mating bearing surfaces of the locking member are defined by a channel formed in the locking member.

11. The chaff pan assembly of claim 7, wherein the first bearing surface of the three different bearing surfaces comprises a rod that is fixed to the one arm.

12. The chaff pan assembly of claim 1, wherein the one arm is pivotably connected to the stationary bracket by a pivot joint.

13. The chaff pan assembly of claim 1, further comprising a user handle on the locking member.

14. An agricultural vehicle comprising the chaff pan assembly of claim 1.

15. The agricultural vehicle of claim 14, wherein the agricultural vehicle is a combine harvester.

16. The chaff pan assembly of claim 1, wherein the locking member moves independently of the stationary bracket and the linkage.
